

v1.0Neutron Deep Dive_ One-Nucleon Universe

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Nucleon Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026
Applies to: v1.0Neutron Deep Dive_ One-Nucleon Universe V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Neutron lifetime puzzle	$\tau \sim 880$ s, bottle/beam 3σ	UCN τ 2023 + beam: 3.4σ persists. Dark neutron decay $n \rightarrow \chi \gamma$ at BR~1% proposed. V2.0: configurable dark decay BR	HIGH	Unexplained discrepancy in core neutron lifetime; may indicate BSM physics
Neutron EDM	$nEDM < 1.8 \times 10^{-26}$ e·cm	V2.0: $\theta_{QCD} < 10^{-10}$ confirmed. Axion solution couples to neutron worldline CP phase	MEDIUM	Axion-neutron coupling creates new worldline interaction missing from V1.0
Neutron star EOS	Not in V1.0 scope	GW230529 (LIGO O4): $R_{NS} = 11.7 \pm 1.0$ km, $\Lambda = 190 \pm 120$. NEUTRON_STAR mode added	MEDIUM	Dense matter regime available as simulation setting
Quantum gravitational states	Not covered	GRANIT 2023: $E_n = (m^{2g^{2\hbar}} 2/2)^{1/3} \cdot z_n$ — first 3 states measured. GRAVITY_QUANTUM mode added	LOW	Novel quantum gravity observable now available for engine demonstration

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Neutron
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
-

Unchanged Architecture

All foundational philosophy is **carried forward unchanged:** - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Neutron Deep Dive_ One-Nucleon Universe V1.0 archived | V2.0 is the active specification as of June 2026